Week 2 – Machine Learning Classifier Report

Dataset Description

The Breast Cancer Wisconsin dataset was used for this classification task. The dataset contains 569 samples and 30 numerical features describing characteristics of cell nuclei obtained from breast mass images. The target variable consists of two classes: malignant and benign tumours.

Data preprocessing included checking for missing values, analysing class balance, feature scaling for Logistic Regression, and correlation-based feature selection. Features with an absolute correlation greater than 0.95 were removed, reducing the number of features from 30 to 23.

The machine learning models were implemented using the Scikit-learn framework [1].

Q1. What is the accuracy of your classifier on the test dataset?

Two classification algorithms were trained and evaluated.

Model	Accuracy
Logistic Regression	97.37%
Random Forest	96.49%

Logistic Regression achieved the highest accuracy and therefore emerged as the best-performing classifier.

Q2. Present a full classification report for your best model and interpret each value.

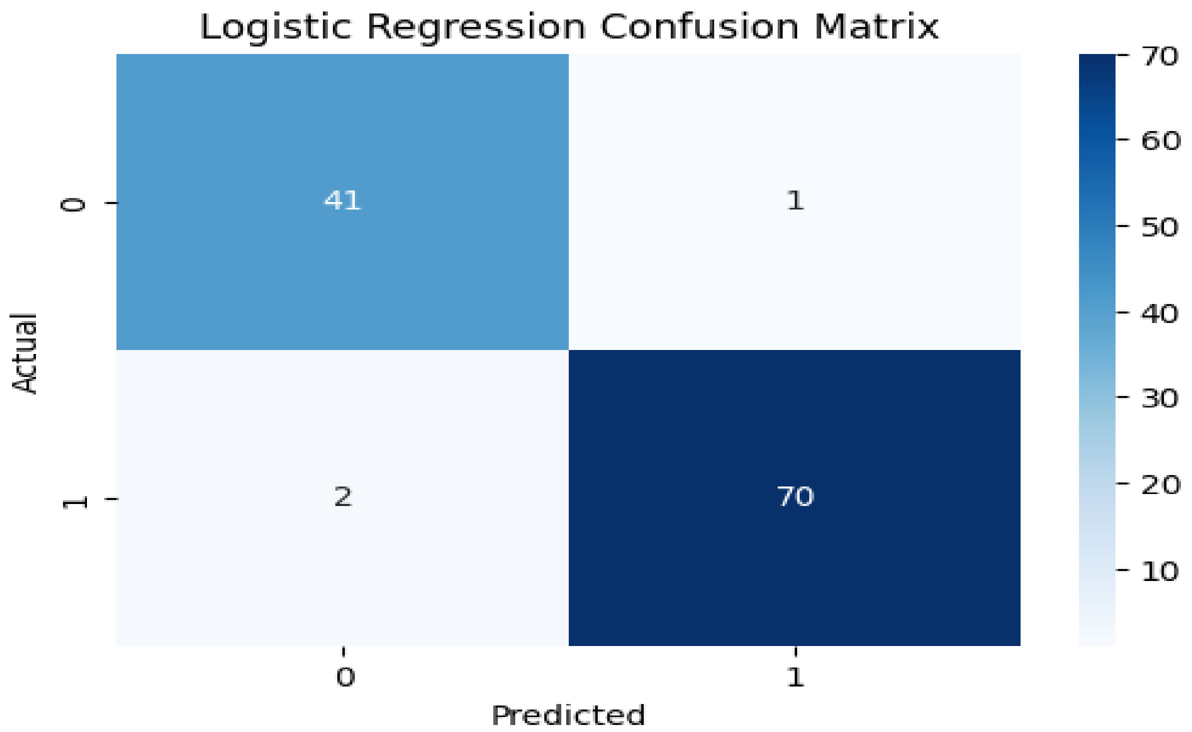
The best model was Logistic Regression.

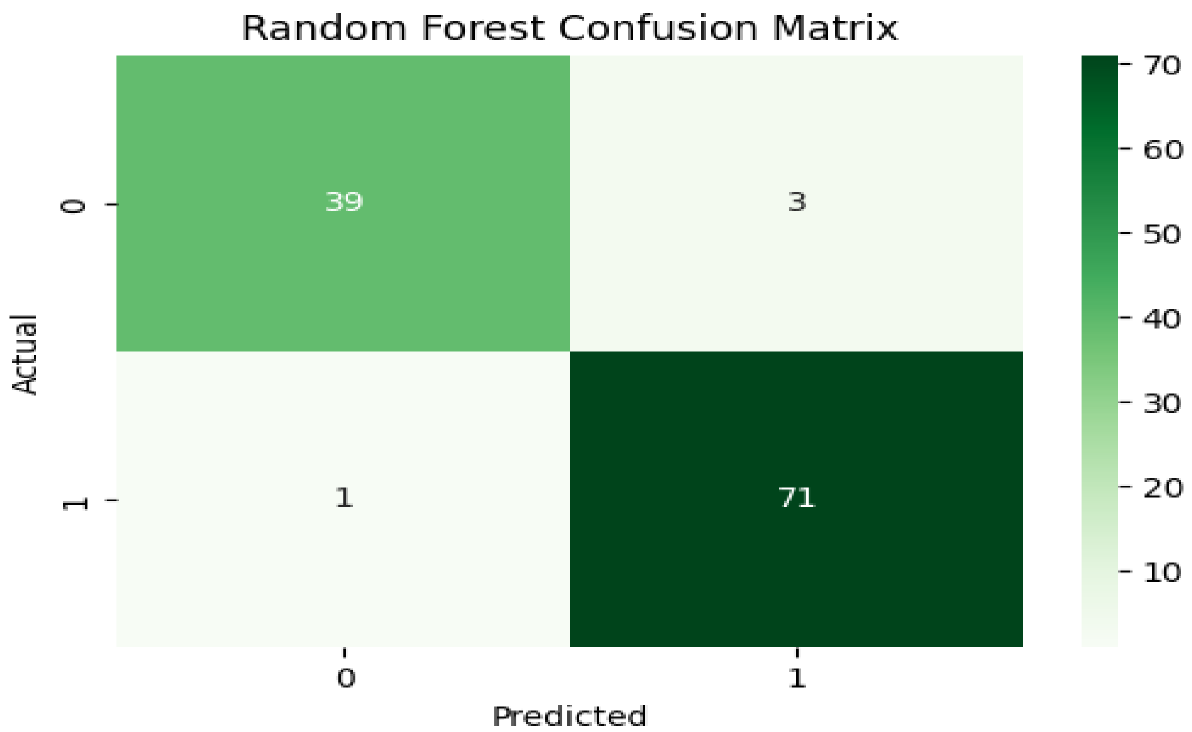
Class	Precision	Recall	F1-Score	Support
Malignant (0)	0.95	0.98	0.96	42
Benign (1)	0.99	0.97	0.98	72

Interpretation

- Precision measures how many predicted samples of a class are actually correct.
- Recall measures how many actual samples of a class are successfully identified.
- F1-score balances precision and recall.
- Support represents the number of samples belonging to each class.

The model achieved a recall of 98% for malignant tumours, which is especially important in medical diagnosis because it minimizes the risk of missing cancer cases.





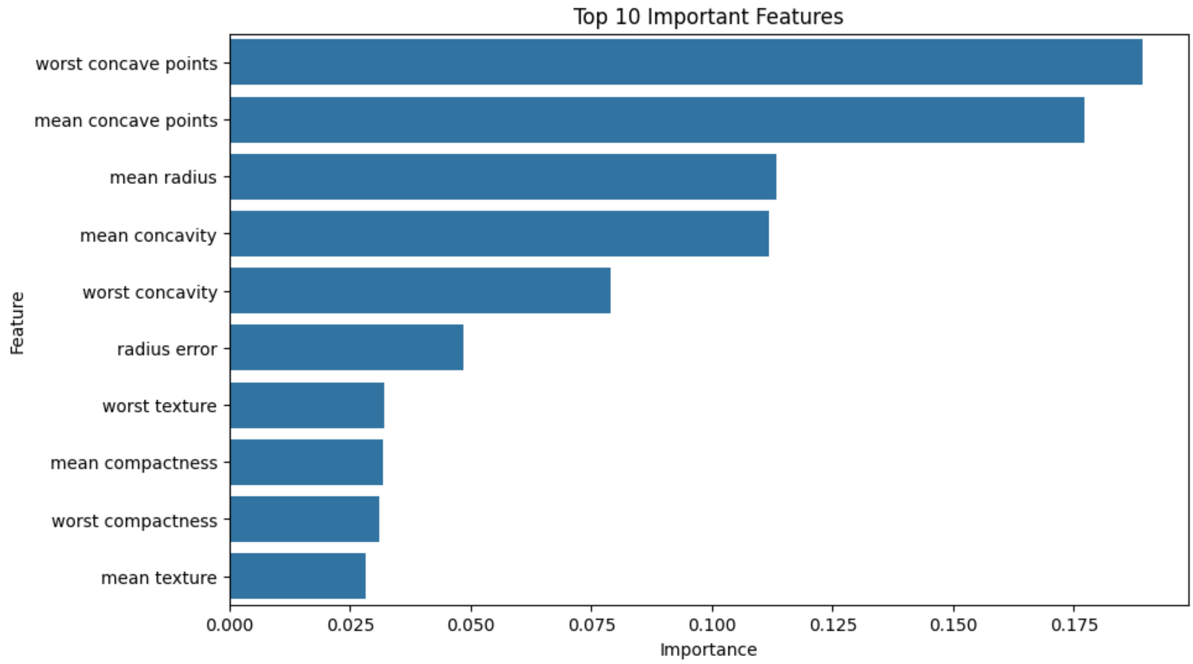
Q3. Which features have the highest predictive importance in your classifier?

Feature importance analysis using Random Forest identified the following features as the most influential:

1. Worst Concave Points
2. Mean Concave Points
3. Mean Radius
4. Mean Concavity
5. Worst Concavity

The feature importance bar chart included in the notebook shows that these features contributed most strongly to classification decisions.

These features describe tumour shape, boundary irregularity, and size, which are medically relevant indicators of malignancy.



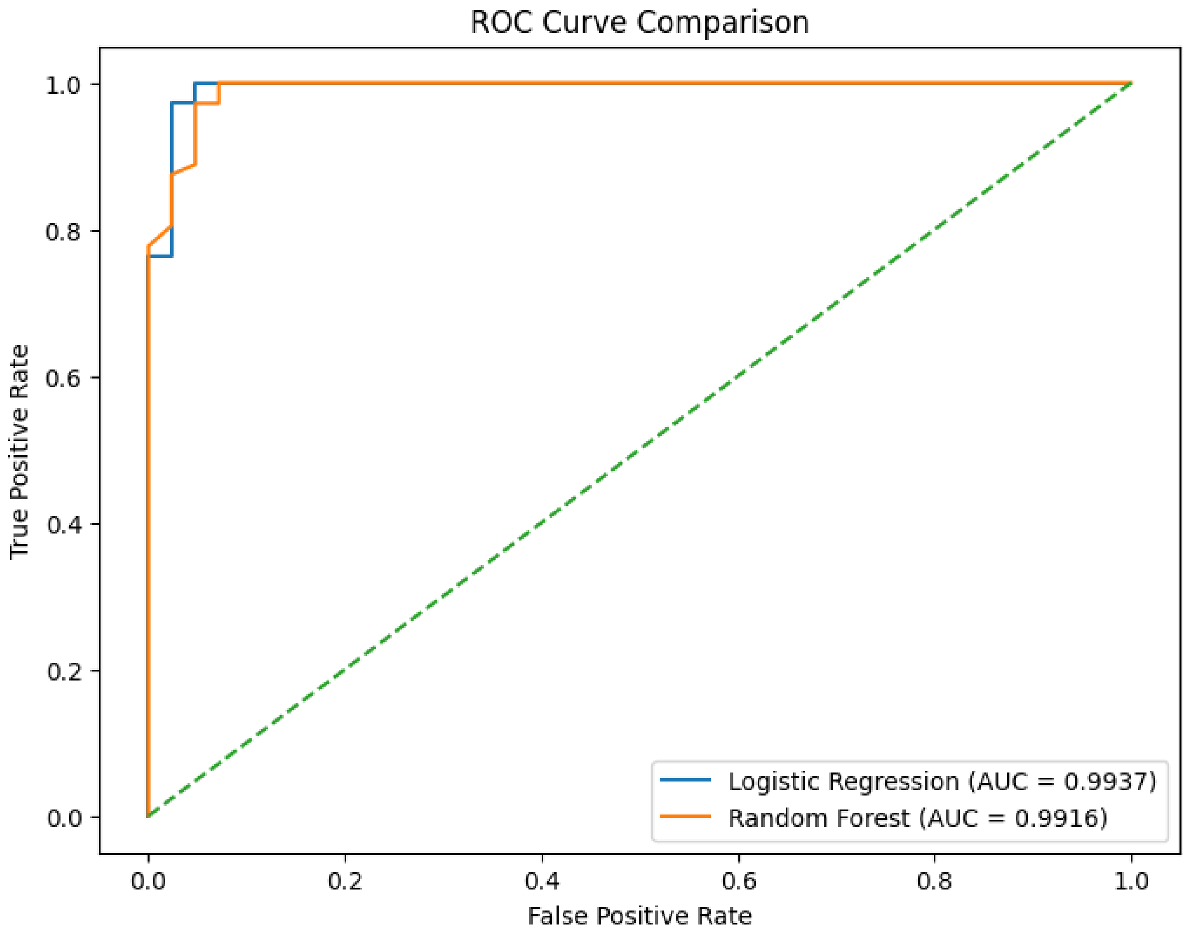
Q4. Train at least two algorithms on the same dataset. Which model performed best and why?

The following comparison was obtained:

Metric	Logistic Regression	Random Forest
Accuracy	97.37%	96.49%
Precision	98.59%	95.95%
Recall	97.22%	98.61%
F1 Score	97.90%	97.26%
AUC	99.37%	99.16%

Metric	Logistic Regression	Random Forest
Cross-Validation Accuracy	97.36%	95.60%

Logistic Regression performed best overall. It achieved higher accuracy, precision, F1-score, AUC, and cross-validation accuracy. Random Forest achieved slightly higher recall for benign tumours, but Logistic Regression demonstrated stronger overall performance and better generalization.



Q5. Do the most important features align with your intuition about the problem domain?

Yes.

The most important features were related to tumour size and boundary irregularity, such as concavity, concave points, and radius. These characteristics are commonly associated with malignant tumours because cancerous growths often exhibit irregular shapes and abnormal cell structures.

The results therefore align well with medical intuition and suggest that the model is learning meaningful biological patterns rather than relying on random correlations.

Q6. How does model performance change across algorithms? What factors might explain the differences?

Although both models achieved excellent performance, Logistic Regression consistently outperformed Random Forest on most evaluation metrics.

Several factors may explain this:

1. The dataset is relatively small (569 samples).
2. The classes are reasonably balanced.
3. The features exhibit strong predictive power and clear separation.
4. The classification boundary appears to be approximately linear.

Because of these characteristics, the simpler Logistic Regression model was able to generalize effectively without requiring the additional complexity of Random Forest.

Cross-validation results further confirmed that Logistic Regression produced more stable performance across different data partitions.

Q7. Apart from accuracy, identify and explain three other metrics used.

Precision

Precision measures the proportion of positive predictions that are correct.

Example:
In medical diagnosis, low precision could result in many healthy patients being incorrectly identified as having cancer.

Recall

Recall measures the proportion of actual positive cases that are successfully detected.

Example:
In cancer screening, recall is often more important than accuracy because missing a cancer case may have severe consequences.

AUC-ROC

AUC measures the ability of a classifier to distinguish between classes across all classification thresholds [3].

Example:
AUC is especially important when comparing multiple classifiers because it evaluates overall discrimination ability rather than performance at a single threshold.

Accuracy alone may be insufficient when classes are imbalanced; therefore precision, recall, and AUC provide a more complete evaluation of model performance [5].

Reflection (R1)

If this model were deployed in a real-world medical application, several ethical concerns would arise.

First, incorrect predictions could affect patient outcomes. False negatives are particularly concerning because a malignant tumour could be incorrectly classified as benign, delaying diagnosis and treatment.

Second, fairness and bias must be considered. A model trained on data that is not representative of the target population may perform poorly for certain demographic groups.

Third, transparency and explainability are important. Healthcare professionals should understand why a prediction was made before relying on it in clinical decision-making.

To address these concerns, the model should undergo rigorous validation, continuous monitoring, and human oversight. Explainable AI techniques and responsible deployment practices should also be adopted. These recommendations align with Google's Responsible AI principles, which emphasize fairness, reliability, safety, privacy, accountability, and transparency [6].

References

[1] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.

[3] T. Fawcett, "An Introduction to ROC Analysis," Pattern Recognition Letters, vol. 27, no. 8, pp. 861–874, 2006.

[5] Google Developers, "Machine Learning Crash Course — Classification," 2024.

[6] Google AI, "Responsible AI Practices," 2024.